

1 Topological insulators and the quantum Hall effect

The first part of these notes follows [Kaufmann et al. \(2016\)](#); for reviews see [Hasan and Kane 2010](#); [Qi and Zhang 2011](#) and the book of [Bernevig and Hughes \(2013\)](#).

A *topological insulator* is an insulator in the bulk with conducting edge (2D) or surface (3D) states on the boundary. More precisely, topological insulators are gapped electronic systems that show topologically protected, nontrivial phases in the presence of time-reversal symmetry. Because of the odd time-reversal symmetry ($\Theta^2 = -1$ for spin- $\frac{1}{2}$ electrons), time-reversal invariant topological insulators are characterized by a $\mathbb{Z}_2$ -valued invariant.

Integer quantum Hall effect. The integer quantum Hall effect (IQHE) is the simplest example of a topological phase. It requires no symmetry other than charge conservation ($U(1)$), time-reversal symmetry is broken by the magnetic field, and the phases are characterized by a $\mathbb{Z}$ -valued invariant.

In the classical Hall setup a magnetic field B is applied perpendicular to a plane of finite size, an electric field E_x is applied, and a Hall current j_y flows in the perpendicular direction. The Hall conductance is

$$\sigma_{xy} = \frac{j_y}{E_x} = \nu \frac{e^2}{h}, \quad (1.1)$$

with ν the filling factor. When the magnetic field is strong and the temperature is low, σ_{xy} is quantized ([von Klitzing et al., 1980](#)),

$$\boxed{\sigma_{xy} = n \frac{e^2}{h}, \quad n \in \mathbb{Z}.} \quad (1.2)$$

The integer n (the TKNN integer) is the first Chern number of the Berry curvature over the Brillouin torus ([Thouless et al., 1982](#)). For still stronger fields and lower temperatures, the fractional quantum Hall effect is observed, in which ν takes fractional values.

2 Landau levels and edge states

The Hamiltonian of a free electron (charge $-e$) in a magnetic field is

$$\mathcal{H} = \frac{1}{2m} (\mathbf{p} + e\mathbf{A})^2 = \frac{\boldsymbol{\pi} \cdot \boldsymbol{\pi}}{2m}, \quad \boldsymbol{\pi} = \mathbf{p} + e\mathbf{A}, \quad (2.1)$$

where $\boldsymbol{\pi}$ is the kinetic momentum (in terms of the covariant derivative, $\pi_\mu = -i\hbar D_\mu$, $D_\mu = \partial_\mu + \frac{ie}{\hbar} A_\mu$). Take $\mathbf{B} = (0, 0, B) = \nabla \times \mathbf{A}$ and the Landau gauge $\mathbf{A} = (0, Bx, 0)$. Then

$$\mathcal{H} = \frac{1}{2m} [p_x^2 + (p_y + eBx)^2]. \quad (2.2)$$

Since $[\mathcal{H}, p_y] = 0$, $\mathcal{H}$ and p_y have a common set of eigenstates and p_y can be replaced by its eigenvalue $\hbar k_y$. With the cyclotron frequency $\omega_c = eB/m$,

$$\mathcal{H} = \frac{p_x^2}{2m} + \frac{1}{2} m \omega_c^2 (x + x_0)^2, \quad x_0 = \frac{\hbar k_y}{eB} = \frac{\hbar k_y}{m \omega_c}, \quad (2.3)$$

a harmonic oscillator centred at $-x_0$. The spectrum is

$$E_n = \hbar \omega_c \left(n + \frac{1}{2} \right), \quad n = 0, 1, 2, \dots, \quad (2.4)$$

the n th Landau level, with eigenfunctions

$$\psi_{n,k_y}(x, y) = e^{ik_y y} \phi_n(x + x_0), \quad (2.5)$$

where ϕ_n are the harmonic-oscillator eigenfunctions. In the bulk the energy does not depend on k_y ; near a boundary in x the confining potential lifts the Landau levels, and $E_n(k_y)$ acquires a dispersion: these are the edge states.

Edge Hamiltonian. More generally, let the bulk Hamiltonian $\mathcal{H}_b(x, y)$ be translation invariant along y . An edge Hamiltonian is obtained from it by a partial Fourier transform in y : the momentum k_y is treated as a parameter, and the edge Hamiltonian is a one-parameter family $\mathcal{H}_e(x, k_y)$ with

$$\mathcal{H}_e(x, k_y) \phi_n(x, k_y) = E_n(k_y) \phi_n(x, k_y). \quad (2.6)$$

3 The Bloch bundle

Thouless, Kohmoto, Nightingale and den Nijs (Thouless et al., 1982) showed that the integer n in $\sigma_{xy} = ne^2/h$ can be identified with the first Chern number of a vector bundle over the Brillouin torus (Simon, 1983).

For the 2D square lattice $\Gamma \simeq \mathbb{Z}^2 \subset \mathbb{R}^2$, the Pontryagin dual (the group of characters $e^{i\mathbf{k} \cdot \mathbf{R}}$, which is how Fourier transformation generalizes to locally compact abelian groups) is the 2-torus $\mathbb{T}^2$. This space, dual to the lattice, is the Brillouin zone, i.e. the space of crystal momenta $\mathbf{k}$; with periodic identification it is the Brillouin torus.

The electronic states in a crystal are Bloch functions,

$$\psi_{\mathbf{k}}(\mathbf{r}) = e^{i\mathbf{k} \cdot \mathbf{r}} u_{\mathbf{k}}(\mathbf{r}), \quad u_{\mathbf{k}}(\mathbf{r} + \mathbf{R}) = u_{\mathbf{k}}(\mathbf{r}) \quad \forall \mathbf{R} \in \Gamma, \quad (3.1)$$

and the periodic parts satisfy

$$\mathcal{H}(\mathbf{k}) u_n(\mathbf{k}) = E_n(\mathbf{k}) u_n(\mathbf{k}), \quad \mathcal{H}(\mathbf{k}) = e^{-i\mathbf{k} \cdot \mathbf{r}} \mathcal{H} e^{i\mathbf{k} \cdot \mathbf{r}}, \quad (3.2)$$

for the n th band, assumed non-degenerate (no band crossing). For each n , $u_n(\mathbf{k})$ spans a subspace $\mathcal{H}_n$ and $\mathcal{H} = \bigoplus_n \mathcal{H}_n$. We write $|u_n(\mathbf{k})\rangle \equiv |n, \mathbf{k}\rangle$, $\langle n, \mathbf{k} | n, \mathbf{k} \rangle = 1$.

Mathematically, the $u_n(\mathbf{k})$ are sections of a Hilbert bundle $E \rightarrow B$ whose fibre over $\mathbf{k} \in B$ is spanned by $u_n(\mathbf{k})$; the base can be any generalized Brillouin zone, e.g. S^d or $\mathbb{T}^d$. For the n th occupied band ($E_n < E_F$) one has a rank-1 (line) bundle

$$\mathcal{L}_n \rightarrow \mathbb{T}^2, \quad \mathbf{k} \mapsto |u_n(\mathbf{k})\rangle. \quad (3.3)$$

If N bands are occupied, the *Bloch bundle* is the rank- N bundle given by the Whitney sum

$$\mathcal{B} = \bigoplus_{n=1}^N \mathcal{L}_n \rightarrow \mathbb{T}^2. \quad (3.4)$$

Its space of sections is the physical Hilbert space of occupied states; the sections themselves are the Bloch wavefunctions.

Bloch Hamiltonian in a magnetic field. For the QHE on a lattice, in the Landau gauge $\mathbf{A} = (0, eBx)$ (units with the charge absorbed),

$$\mathcal{H}_{\text{Bloch}}(k_1, k_2) = \frac{1}{2m} (\hat{\mathbf{p}}_{\text{Bloch}} + e\mathbf{A})^2 + U(x, y), \quad \hat{\mathbf{p}}_{\text{Bloch}} = \left(-i\hbar\partial_x + \hbar k_1, -i\hbar\partial_y + \hbar k_2 \right), \quad (3.5)$$

with the periodic potential

$$U(x, y) = U_1 \cos(2\pi x/a) + U_2 \cos(2\pi y/b). \quad (3.6)$$

For rational flux p/q (flux quanta per unit cell) the magnetic translations close on a $q \times 1$ magnetic unit cell, and the boundary conditions become

$$u_{k_1 k_2}(x + qa, y) e^{-2\pi i p y/b} = u_{k_1 k_2}(x, y + b) = u_{k_1 k_2}(x, y). \quad (3.7)$$

4 Berry phase, connection and curvature

Berry phase

Consider a quantum system going through a cyclic adiabatic evolution (Berry, 1984). The Hamiltonian $\mathcal{H}(\mathbf{R})$ depends on parameters $\mathbf{R}(t)$, with instantaneous eigenvalues $E_n(\mathbf{R})$ and eigenstates $|n(\mathbf{R})\rangle$, and the initial state is $|n(\mathbf{R}(0))\rangle$. At time t ,

$$|\psi_n(t)\rangle = e^{i\gamma_n(t)} e^{-\frac{i}{\hbar} \int_0^t dt' E_n(\mathbf{R}(t'))} |n(\mathbf{R}(t))\rangle, \quad (4.1)$$

where the second factor is the dynamical phase and γ_n the geometric phase. Inserting into the Schrödinger equation,

$$\gamma_n(t) = i \int_0^t dt' \langle n(\mathbf{R}(t')) | \frac{d}{dt'} | n(\mathbf{R}(t')) \rangle = i \int_{\mathbf{R}(0)}^{\mathbf{R}(t)} d\mathbf{R} \cdot \langle n(\mathbf{R}) | \nabla_{\mathbf{R}} | n(\mathbf{R}) \rangle. \quad (4.2)$$

It depends only on the path in parameter space, not on how fast it is traversed. For a closed path C , $\mathbf{R}(T) = \mathbf{R}(0)$,

$$\gamma_n = i \oint_C d\mathbf{R} \cdot \langle n(\mathbf{R}) | \nabla_{\mathbf{R}} | n(\mathbf{R}) \rangle. \quad (4.3)$$

Gauge transformation

Under $|n'(\mathbf{R})\rangle = e^{-i\beta(\mathbf{R})} |n(\mathbf{R})\rangle$ the Berry phase of an open path changes as

$$\gamma'_n(t) = \gamma_n(t) + \beta(\mathbf{R}(t)) - \beta(\mathbf{R}(0)). \quad (4.4)$$

For a closed path single-valuedness of $|n'\rangle$ requires $\beta(\mathbf{R}(T)) - \beta(\mathbf{R}(0)) = 2\pi m$, $m \in \mathbb{Z}$, hence $\gamma'_n = \gamma_n \pmod{2\pi}$: the Berry phase of a closed loop is physical.

Berry connection and curvature

The *Berry connection*

$$\mathbf{A}_n(\mathbf{R}) = i \langle n(\mathbf{R}) | \nabla_{\mathbf{R}} | n(\mathbf{R}) \rangle \quad (4.5)$$

is not a physical quantity: under the gauge transformation above $\mathbf{A}'_n = \mathbf{A}_n + \nabla_{\mathbf{R}}\beta$. The Berry phase is its line integral, $\gamma_n = \oint_C d\mathbf{R} \cdot \mathbf{A}_n$. The *Berry curvature*, a rank-2 antisymmetric tensor,

$$F_{\mu\nu}^n(\mathbf{R}) = \partial_{R^\mu} A_\nu^n - \partial_{R^\nu} A_\mu^n \quad (\text{in 3D: } \mathbf{F}_n = \nabla_{\mathbf{R}} \times \mathbf{A}_n), \quad (4.6)$$

is gauge invariant. The analogy with electromagnetism is

Berry connection $\mathbf{A}_n$	vector potential (gauge field)
Berry curvature $F_{\mu\nu}^n$	field-strength tensor (gauge invariant)
Berry phase γ_n	Aharonov–Bohm phase (flux through C)

By Stokes' theorem, $\gamma_n = \int_S d\mathbf{S} \cdot \mathbf{F}_n$ for a surface S with $\partial S = C$.

Chern number

If S is a closed manifold the boundary term vanishes, but γ_n is still defined only modulo 2π . Consequently (Chern theorem) the integral of the Berry curvature over a closed manifold is quantized in units of 2π . For the Bloch bundle, with connection $a^n(\mathbf{k}) = i \langle n, \mathbf{k} | d | n, \mathbf{k} \rangle$ and curvature $f^n = da^n$,

$$c_1^n = \frac{1}{2\pi} \int_{\mathbb{T}^2} f^n = \frac{1}{2\pi} \int_{\mathbb{T}^2} d^2k \left(\frac{\partial a_y^n}{\partial k_x} - \frac{\partial a_x^n}{\partial k_y} \right) \in \mathbb{Z}, \quad (4.7)$$

and the first Chern number of the Bloch bundle $\mathcal{B} \rightarrow \mathbb{T}^2$ is

$$c_1 = \sum_{n=1}^N c_1^n \in \mathbb{Z}. \quad (4.8)$$

The Hall conductance is $\sigma_{xy} = -c_1 e^2/h$ (the overall sign depends on the sign conventions for e and B) (Thouless et al., 1982).

5 Two-band Dirac Hamiltonians

The quantum Hall system can be viewed as a topological insulator whose effective Hamiltonian is of Dirac type. Consider a two-band model defined by a map

$$\mathbf{h} : \mathbb{T}^2 \rightarrow \mathbb{R}^3, \quad \mathbf{k} = (k_x, k_y) \mapsto \mathbf{h}(\mathbf{k}), \quad (5.1)$$

and the Hamiltonian

$$\mathcal{H}(\mathbf{k}) = h_x(\mathbf{k})\sigma_x + h_y(\mathbf{k})\sigma_y + h_z(\mathbf{k})\sigma_z = \mathbf{h}(\mathbf{k}) \cdot \boldsymbol{\sigma}, \quad E_{\pm}(\mathbf{k}) = \pm |\mathbf{h}(\mathbf{k})|. \quad (5.2)$$

The parent Hamiltonian $\mathcal{H} = \mathbf{R} \cdot \boldsymbol{\sigma}$ on $\mathbf{R} \in \mathbb{R}^3$ is non-degenerate except at $\mathbf{R} = 0$. The degeneracy point acts as a monopole of Berry curvature: for the lower band $\mathbf{F}_- = +\frac{1}{2}\mathbf{R}/R^2$, so that the flux through any sphere around $\mathbf{R} = 0$ is 2π ; this monopole is the source of the Chern numbers.

Winding number. If $\mathbf{h}(\mathbf{k}) \neq 0$ everywhere on $\mathbb{T}^2$ (the system is gapped), define the unit vector

$$\hat{\mathbf{d}} : \mathbb{T}^2 \rightarrow S^2, \quad \hat{\mathbf{d}}(\mathbf{k}) = \frac{\mathbf{h}(\mathbf{k})}{|\mathbf{h}(\mathbf{k})|}. \quad (5.3)$$

The Chern number of the lower band equals the winding number (degree) of $\hat{\mathbf{d}}$, i.e. the number of times $\hat{\mathbf{d}}$ covers the sphere:

$$c_1 = \frac{1}{4\pi} \int_{\mathbb{T}^2} d^2k \hat{\mathbf{d}} \cdot (\partial_{k_x} \hat{\mathbf{d}} \times \partial_{k_y} \hat{\mathbf{d}}). \quad (5.4)$$

Generalization to N -band models. Replacing $\sigma/2$ by spin- S matrices $\mathbf{S}$ with $S = (N - 1)/2$, $\mathcal{H} = 2\mathbf{h} \cdot \mathbf{S}$ has bands $E_m = 2m|\mathbf{h}|$, $m = -S, \dots, S$, and the band labelled by m has Chern number $c_1^{(m)} = -2mW$, where W is the winding number (5.4). For $S = \frac{1}{2}$ this gives $\pm W$.

A lattice model

A particular choice of $\mathbf{h}(\mathbf{k})$ is

$$\mathcal{H}(\mathbf{k}) = A \sin k_x \sigma_x + A \sin k_y \sigma_y + \left[M - 4B \left(\sin^2 \frac{k_x}{2} + \sin^2 \frac{k_y}{2} \right) \right] \sigma_z, \quad (5.5)$$

where A, B, M are parameters (M is the mass parameter). It describes electrons on a 2D square lattice with spin-orbit coupling (the spin-up block of the Bernevig–Hughes–Zhang model (Bernevig et al., 2006)). h_z takes the values $M, M - 4B, M - 4B, M - 8B$ at the four time-reversal invariant momenta $(0, 0), (\pi, 0), (0, \pi), (\pi, \pi)$, where $h_x = h_y = 0$. The gap closes when one of these vanishes, and between these points the Chern number is constant. Evaluating Eq. (5.4) (for $A > 0$):

range of M/B	c_1 (lower band)
$M/B < 0$ or $M/B > 8$	0
$0 < M/B < 4$	$\text{sgn } B$
$4 < M/B < 8$	$-\text{sgn } B$

Continuum limit. Near $\mathbf{k} = 0$, with $\sin k \simeq k$ and $4 \sin^2(k/2) \simeq k^2$,

$$\mathcal{H}(\mathbf{k}) = A \mathbf{k} \cdot \boldsymbol{\sigma} + (M - B|\mathbf{k}|^2) \sigma_z, \quad (5.6)$$

a massive Dirac Hamiltonian with a quadratic correction $B|\mathbf{k}|^2 \sigma_z$. Then $\hat{\mathbf{d}}(\mathbf{k} = 0) = \text{sgn}(M)\hat{z}$ and $\hat{\mathbf{d}}(|\mathbf{k}| \rightarrow \infty) = -\text{sgn}(B)\hat{z}$, and

$$c_1 = \frac{1}{2} [\text{sgn}(M) + \text{sgn}(B)]. \quad (5.7)$$

The phase is topological ($c_1 = \pm 1$) for $MB > 0$ and trivial ($c_1 = 0$) for $MB < 0$. The transition at $M = 0$ is accompanied by the closing of the gap at $\mathbf{k} = 0$. Without the $B|\mathbf{k}|^2$ term, a single Dirac cone gives the half-integer value $c_1 = \frac{1}{2} \text{sgn } M$: the regularization at large $|\mathbf{k}|$ is needed for an integer.

6 Bulk–edge correspondence

The first Chern number characterizes a bulk state of the QHE. At an interface with an insulator of different Chern number, edge states appear (Halperin, 1982; Hatsugai, 1993). Consider two copies of the model with different ground-state Chern numbers next to each other along an interface: a domain wall. The aim is to connect the edge states and the bulk through an index theorem.

Consider a two-band system with a gap in which the Fermi level lies. The generalized index theorem for such Dirac-type Hamiltonians connects the analytical index of the edge states (their spectral flow) with the topological index of the bulk (the first Chern number): this is the *bulk–edge correspondence*.

Dirac Hamiltonian in position space. The bulk Hamiltonian is

$$\mathcal{H}_b = -i\partial_x\sigma_x - i\partial_y\sigma_y + (M + B\partial_x^2 + B\partial_y^2)\sigma_z \quad (6.1)$$

($A = 1$). The edge Hamiltonian is obtained by partial Fourier transform in y ,

$$\mathcal{H}_e(x, k_y) = -i\partial_x\sigma_x + k_y\sigma_y + (M + B\partial_x^2 - Bk_y^2)\sigma_z. \quad (6.2)$$

Introduce a domain wall at $x = 0$ separating two bulk systems by letting the mass depend on x ,

$$\lim_{x \rightarrow \pm\infty} M(x) = M_{\pm}. \quad (6.3)$$

The edge states satisfy $\mathcal{H}_e(x, k_y) \phi_n(x, k_y) = E_n(k_y) \phi_n(x, k_y)$. As $|k_y|$ becomes large, $E_n(k_y)$ enters the conduction or valence band (the usual assumption). $\phi_n(x, k_y)$ is called an *edge state* if it connects the valence and conduction bands, i.e. if as k_y is varied $E_n(k_y)$ reaches both bands. Every edge state therefore crosses the Fermi level $E_F = 0$.

Explicit solution (Jackiw–Rebbi). For $B = 0$ the edge state can be found exactly (Jackiw and Rebbi, 1976). Try $\phi(x, k_y) = f(x)\chi$ with a constant spinor χ , $\sigma_y\chi = s\chi$, $s = \pm 1$. The terms $-i\partial_x\sigma_x + M(x)\sigma_z$ annihilate ϕ if

$$-i\sigma_x f'(x)\chi + M(x)\sigma_z f(x)\chi = 0 \Leftrightarrow f'(x)\chi = -M(x)(i\sigma_x\sigma_z)f(x)\chi = -sM(x)f(x)\chi, \quad (6.4)$$

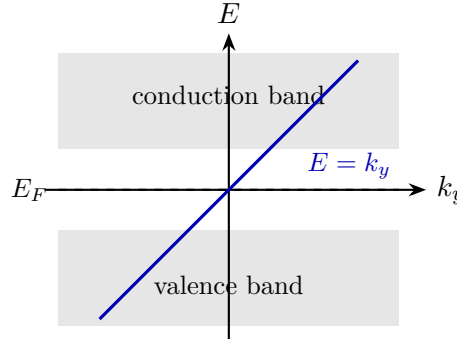
using $i\sigma_x\sigma_z = \sigma_y$. Hence

$$f(x) = \exp\left(-s \int_0^x M(x') dx'\right), \quad (6.5)$$

which is normalizable only if M changes sign: for $M_- < 0 < M_+$ we need $s = +1$. The remaining term gives

$$\mathcal{H}_e\phi = k_y\sigma_y\phi = s k_y \phi \Rightarrow E(k_y) = s k_y. \quad (6.6)$$

This is a single chiral edge mode, crossing $E_F = 0$ once and connecting the valence and conduction bands. Its chirality is set by the sign change of the mass, i.e. by the change of the Chern number across the wall, $c_1(M_+) - c_1(M_-) = \pm 1$ in the continuum model.



7 Spontaneous symmetry breaking

The second part of these notes treats gauge theories, following Rubakov (2002); see also Polyakov (1987), Coleman (1985) and Peskin and Schroeder (1995). We first need the basics of spontaneous symmetry breaking (SSB).

Discrete symmetry

Consider a real scalar field in 3 + 1 dimensions,

$$\mathcal{L} = \frac{1}{2}(\partial_\mu \varphi)^2 - \frac{m^2}{2}\varphi^2 - \frac{\lambda}{4}\varphi^4 \equiv T - V, \quad (7.1)$$

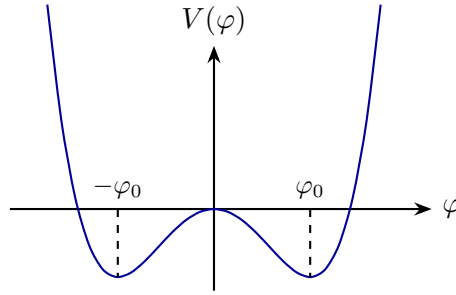
with the symmetry $\varphi(x) \rightarrow -\varphi(x)$. The energy functional is

$$E = \int d^3x \left[\frac{1}{2}(\partial_0 \varphi)^2 + \frac{1}{2}(\partial_i \varphi)^2 + \frac{m^2}{2}\varphi^2 + \frac{\lambda}{4}\varphi^4 \right], \quad V(\varphi) = \frac{m^2}{2}\varphi^2 + \frac{\lambda}{4}\varphi^4. \quad (7.2)$$

For a stable vacuum the potential must be bounded from below, so $\lambda > 0$. The ground state is the field configuration of minimal energy. The minimum is reached for fields that are constant in time, $\partial_0 \varphi = 0$, and homogeneous in space, $\partial_i \varphi = 0$, so we only need to minimize $V(\varphi)$:

- (1) $m^2 > 0$: the minimum is at $\varphi = 0$. The ground state does not break the symmetry of the model.
- (2) $m^2 < 0$: write $\mu^2 = -m^2 > 0$. Now $\varphi = 0$ is a local maximum, and the ground states are

$$\varphi = \pm \varphi_0, \quad \varphi_0 = \frac{\mu}{\sqrt{\lambda}}. \quad (7.3)$$



$V(\varphi)$ is the energy density of a homogeneous field; the potential energy of a homogeneous configuration is $\Omega V(\varphi)$, with Ω the volume of space. To transfer the field from one ground state to the other one has to inject an energy proportional to Ω . Thus for a system with large volume, $\Omega \rightarrow \infty$, we have to choose one of the ground states, say $\varphi = +\varphi_0$, and consider perturbations around it. The ground state is not invariant under $\varphi \rightarrow -\varphi$: the symmetry is broken spontaneously. Its energy is $E_0 = \Omega V(\varphi_0) = -\Omega \mu^4/(4\lambda)$. Adding this constant to $\mathcal{L}$,

$$\mathcal{L} = \frac{1}{2}(\partial_\mu \varphi)^2 - \frac{\lambda}{4}(\varphi^2 - \varphi_0^2)^2. \quad (7.4)$$

For fluctuations $\varphi(x) = \varphi_0 + \chi(x)$,

$$\mathcal{L}_\chi = \frac{1}{2}(\partial_\mu \chi)^2 - \mu^2 \chi^2 - \sqrt{\lambda} \mu \chi^3 - \frac{\lambda}{4} \chi^4, \quad (7.5)$$

so χ is a particle with mass $m_\chi = \sqrt{2}\mu$, and the cubic term reflects the broken symmetry.

Why no tunnelling. In quantum mechanics with a symmetric double well, the ground state is the symmetric superposition of wavefunctions concentrated near each well, because of tunnelling between the wells. In field theory this tunnelling must occur simultaneously in the whole space. For homogeneous fields the action is

$$S = \Omega \int dt \left(\frac{\dot{\varphi}^2}{2} - V(\varphi) \right), \quad (7.6)$$

the action of a particle of mass Ω in the potential $\mathcal{U}(\varphi) = \Omega V(\varphi)$. The WKB tunnelling amplitude between $\pm\varphi_0$ is

$$\mathcal{A} \propto \exp \left(- \int_{-\varphi_0}^{\varphi_0} \sqrt{2\Omega\mathcal{U}} d\varphi \right) = \exp \left(- \Omega \int_{-\varphi_0}^{\varphi_0} \sqrt{2V(\varphi)} d\varphi \right) \xrightarrow{\Omega \rightarrow \infty} 0. \quad (7.7)$$

Continuous symmetry: global $U(1)$

For a complex field $\varphi = (\varphi_1 + i\varphi_2)/\sqrt{2}$,

$$\mathcal{L} = \partial_\mu \varphi^* \partial^\mu \varphi - V(\varphi^* \varphi), \quad V = -\mu^2 \varphi^* \varphi + \lambda(\varphi^* \varphi)^2 + c, \quad (7.8)$$

invariant under $\varphi \rightarrow e^{i\alpha} \varphi$. The minima form a continuous set (a circle in the (φ_1, φ_2) plane),

$$\varphi = e^{i\alpha} \frac{\varphi_0}{\sqrt{2}}, \quad \varphi_0 = \frac{\mu}{\sqrt{\lambda}}. \quad (7.9)$$

Transitions between different minima can be made without increasing the potential energy, by moving along the circle of minima. They still require an infinite energy for $\Omega \rightarrow \infty$, since the kinetic energy of a homogeneous change is $E_{\text{kin}} = \Omega |\dot{\varphi}|^2$.

Partial symmetry breaking and Goldstone modes

Consider a triplet φ^a , $a = 1, 2, 3$, with

$$\mathcal{L} = \frac{1}{2} \partial_\mu \varphi^a \partial^\mu \varphi^a - V(\varphi), \quad V = -\frac{\mu^2}{2} \varphi^a \varphi^a + \frac{\lambda}{4} (\varphi^a \varphi^a)^2 + \frac{\mu^4}{4\lambda}, \quad (7.10)$$

invariant under $SO(3)$. V is minimal on the sphere $\varphi^a \varphi^a = \varphi_0^2$, $\varphi_0 = \mu/\sqrt{\lambda}$. Choose $\varphi^{(0)} = (0, 0, \varphi_0)$. This vacuum is invariant only under rotations about the 3-axis,

$$\varphi^1 \rightarrow \cos \alpha \varphi^1 - \sin \alpha \varphi^2, \quad \varphi^2 \rightarrow \sin \alpha \varphi^1 + \cos \alpha \varphi^2, \quad \varphi^3 \rightarrow \varphi^3, \quad (7.11)$$

i.e. $SO(3)$ is broken to $SO(2)$. With $\varphi^1 = \theta^1$, $\varphi^2 = \theta^2$, $\varphi^3 = \varphi_0 + \chi$, to quadratic order

$$\mathcal{L} = \frac{1}{2} (\partial_\mu \theta^1)^2 + \frac{1}{2} (\partial_\mu \theta^2)^2 + \frac{1}{2} (\partial_\mu \chi)^2 - \mu^2 \chi^2. \quad (7.12)$$

θ^1, θ^2 are massless: they are the Goldstone modes (Goldstone, 1961). In general, for SSB $G_1 \rightarrow G_2$ the number of Goldstone modes is the number of broken generators,

$$\boxed{n_{\text{GB}} = \dim G_1 - \dim G_2}, \quad (7.13)$$

here $3 - 1 = 2$.

8 Topological solitons: the kink

The discrete case already contains a topological object (Rubakov, 2002; Coleman, 1985). In $1 + 1$ dimensions consider a static field with $\varphi(-\infty) = -\varphi_0$ and $\varphi(+\infty) = +\varphi_0$. It cannot be deformed into a vacuum with finite energy, since that would require changing φ in an infinite region; the configurations are classified by the topological charge

$$Q = \frac{1}{2\varphi_0} [\varphi(+\infty) - \varphi(-\infty)] \in \{0, \pm 1\}. \quad (8.1)$$

The static energy can be written as

$$E = \int dx \left[\frac{1}{2} \varphi'^2 + \frac{\lambda}{4} (\varphi^2 - \varphi_0^2)^2 \right] = \int dx \frac{1}{2} \left[\varphi' - \sqrt{\frac{\lambda}{2}} (\varphi_0^2 - \varphi^2) \right]^2 + \sqrt{\frac{\lambda}{2}} \int_{-\varphi_0}^{\varphi_0} (\varphi_0^2 - \varphi^2) d\varphi. \quad (8.2)$$

The minimum in the sector $Q = 1$ is reached when the square vanishes, $\varphi' = \sqrt{\lambda/2} (\varphi_0^2 - \varphi^2)$, giving the kink

$$\boxed{\varphi_K(x) = \varphi_0 \tanh\left(\frac{\mu x}{\sqrt{2}}\right), \quad E_K = \frac{2\sqrt{2}}{3} \frac{\mu^3}{\lambda}.} \quad (8.3)$$

The kink is the field-theory version of the domain wall between the two vacua; its width is $\sim 1/\mu$ and its energy is $\propto 1/\lambda$, i.e. nonperturbative in the coupling.

9 Counting degrees of freedom of gauge fields

In $3 + 1$ dimensions a gauge field A_μ starts with four components (A_0, A_1, A_2, A_3) . The kinetic term is

$$\mathcal{L} \supset -\frac{1}{4} F_{\mu\nu} F^{\mu\nu}, \quad F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu \quad (\text{abelian case}), \quad (9.1)$$

with $F_{\mu\nu}$ antisymmetric. Hence there is no term $\dot{A}_0 = \partial_0 A_0$: A_0 is not dynamical. It is fully determined by the equation of motion $\nabla \cdot \mathbf{E} = 0$, with $\mathbf{E} = -\nabla A_0 - \partial_t \mathbf{A}$:

$$\nabla^2 A_0 + \nabla \cdot \frac{\partial \mathbf{A}}{\partial t} = 0 \Rightarrow A_0(\mathbf{r}) = \int d^3 x' \frac{(\nabla \cdot \partial_t \mathbf{A})(\mathbf{r}')}{4\pi |\mathbf{r} - \mathbf{r}'|}. \quad (9.2)$$

Thus A_0 is not independent and need not be specified at the initial time, leaving $4 - 1 = 3$ degrees of freedom.

Lorenz gauge. Suppose we start with A'_μ satisfying $\partial_\mu A'^\mu = f(x)$. With $A_\mu = A'_\mu + \partial_\mu \Theta$ we get $\partial_\mu A^\mu = 0$ provided $\partial_\mu \partial^\mu \Theta = -f$, which always has a solution. But we can still add any Θ with $\square \Theta = 0$ (the equation of motion of a massless scalar), which has nontrivial solutions. Using this residual gauge freedom we can also set $\nabla \cdot \mathbf{A} = 0$ (Coulomb gauge); then, in the absence of charges, $A_0 = 0$. The three remaining components satisfy $\nabla \cdot \mathbf{A} = 0$, which leaves

$$\boxed{2 \text{ degrees of freedom for a massless gauge field.}} \quad (9.3)$$

For massive gauge fields there is no gauge transformation, so the second step does not apply, while the first one (non-dynamical A_0) still holds: a massive vector field has 3 degrees of freedom.

Ward identity. The same counting follows from the amplitude for emission of a photon with momentum k , $\mathcal{M}(k) = \epsilon_\mu(k) \mathcal{M}^\mu(k)$, with ϵ_μ the polarization vector. The Ward identity $k_\mu \mathcal{M}^\mu(k) = 0$ allows the shift $\epsilon_\mu \rightarrow \epsilon_\mu + c k_\mu$, which preserves $k \cdot \epsilon = 0$ only because $k^2 = 0$ for a massless photon. This extra freedom is absent for massive vector fields, which therefore have one more degree of freedom than massless ones.

Remark. The residual mode $\square\Theta = 0$ is a massless scalar, of Goldstone type. If it were not removed it would appear inside A_μ as a longitudinal mode, giving 3 degrees of freedom. The Higgs mechanism does precisely this: it injects the scalar degree of freedom of a Goldstone mode into the gauge field and makes it massive.

10 $U(1)$ gauge theory and the Higgs mechanism

Consider the abelian Higgs model (Higgs, 1964; Englert and Brout, 1964),

$$\mathcal{L} = -\frac{1}{4}F_{\mu\nu}F^{\mu\nu} + (D_\mu\varphi)^*(D^\mu\varphi) - \left[-\mu^2\varphi^*\varphi + \lambda(\varphi^*\varphi)^2 \right], \quad (10.1)$$

$$F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu, \quad D_\mu\varphi = (\partial_\mu - ieA_\mu)\varphi, \quad (10.2)$$

with φ a complex scalar field. It is invariant under the gauge transformations

$$A_\mu(x) \rightarrow A_\mu(x) + \frac{1}{e}\partial_\mu\alpha(x), \quad \varphi(x) \rightarrow e^{i\alpha(x)}\varphi(x). \quad (10.3)$$

The energy is

$$E(A_\mu, \varphi) = \int d^3x \left[\frac{1}{2}F_{0i}^2 + \frac{1}{4}F_{ij}^2 + (D_0\varphi)^*(D_0\varphi) + (D_i\varphi)^*(D_i\varphi) + V(\varphi^*\varphi) \right]. \quad (10.4)$$

There is a functional freedom in the choice of ground state: E is gauge invariant, so if $(A_\mu^{(v)}, \varphi^{(v)})$ is a ground state, then so is $(A_\mu^{(v)} + \frac{1}{e}\partial_\mu\alpha, e^{i\alpha}\varphi^{(v)})$ for any $\alpha(x)$.

The first two terms vanish (are minimal) when the electric and magnetic fields vanish, i.e. $A_\mu = \frac{1}{e}\partial_\mu\alpha$ is a pure gauge. The third and fourth terms vanish when

$$D_\mu\varphi = (\partial_\mu - ie\partial_\mu\alpha)\varphi = 0 \Rightarrow \varphi(x) = e^{i\alpha(x)}\frac{\varphi_0}{\sqrt{2}}, \quad (10.5)$$

with φ_0 independent of x , fixed by minimizing the scalar potential: $\varphi_0 = \mu/\sqrt{\lambda}$. Choosing $\alpha = 0$, the vacuum is

$$A_\mu^{(v)} = 0, \quad \varphi^{(v)} = \frac{\varphi_0}{\sqrt{2}}. \quad (10.6)$$

Fluctuations. Write $\varphi(x) = \frac{1}{\sqrt{2}}(\varphi_0 + \chi(x) + i\theta(x))$. Then

$$D_\mu\varphi = \frac{1}{\sqrt{2}}(\partial_\mu\chi + i\partial_\mu\theta - ie\varphi_0 A_\mu) + O(\chi A, \theta A), \quad (10.7)$$

and to quadratic order

$$\mathcal{L} = -\frac{1}{4}F_{\mu\nu}F^{\mu\nu} + \frac{1}{2}(\partial_\mu\chi)^2 - \mu^2\chi^2 + \frac{e^2\varphi_0^2}{2}\left(A_\mu - \frac{1}{e\varphi_0}\partial_\mu\theta\right)^2. \quad (10.8)$$

Define the gauge-invariant field

$$B_\mu = A_\mu - \frac{1}{e\varphi_0}\partial_\mu\theta, \quad B_{\mu\nu} = \partial_\mu B_\nu - \partial_\nu B_\mu = F_{\mu\nu}. \quad (10.9)$$

The Lagrangian becomes

$$\mathcal{L} = -\frac{1}{4}B_{\mu\nu}B^{\mu\nu} + \frac{e^2\varphi_0^2}{2}B_\mu B^\mu + \frac{1}{2}(\partial_\mu\chi)^2 - \mu^2\chi^2. \quad (10.10)$$

- B_μ is a massive vector field with $m_V = e\varphi_0 = e\mu/\sqrt{\lambda}$.
- χ is a massive scalar (the Higgs field) with $m_\chi = \sqrt{2}\mu$.
- θ , the would-be Goldstone mode, has disappeared from the Lagrangian: it has become the longitudinal component of B_μ (choosing $\theta = 0$ is the unitary gauge).

Counting: before, 2 (massless A_μ) + 2 (complex φ) = 4; after, 3 (massive B_μ) + 1 (χ) = 4.

Gauge transformation of B_μ and χ . For infinitesimal α , $A'_\mu = A_\mu + \frac{1}{e}\partial_\mu\alpha$ and $\varphi' \simeq \varphi + i\alpha\varphi$, so

$$\frac{1}{\sqrt{2}}(\varphi_0 + \chi' + i\theta') \simeq \frac{1}{\sqrt{2}}(\varphi_0 + \chi + i\theta) + \frac{i\alpha\varphi_0}{\sqrt{2}}, \quad (10.11)$$

neglecting $\alpha\chi$ and $\alpha\theta$ (second order). Hence

$$\chi' = \chi \quad (\text{gauge invariant}), \quad \theta' = \theta + \alpha\varphi_0 \quad (\text{Goldstone mode shifted}), \quad (10.12)$$

$$B'_\mu = A'_\mu - \frac{1}{e\varphi_0}\partial_\mu\theta' = A_\mu - \frac{1}{e\varphi_0}\partial_\mu\theta = B_\mu \quad (\text{gauge invariant}). \quad (10.13)$$

11 Higgs mechanism in a non-abelian gauge theory

Consider an $SU(2)$ gauge theory with a complex doublet

$$\varphi = \begin{pmatrix} \varphi_1 \\ \varphi_2 \end{pmatrix} \quad (4 \text{ real degrees of freedom}), \quad (11.1)$$

$$\mathcal{L} = -\frac{1}{2}\text{Tr}(G_{\mu\nu}G^{\mu\nu}) + (D_\mu\varphi)^\dagger(D^\mu\varphi) - \left[-\mu^2\varphi^\dagger\varphi + \lambda(\varphi^\dagger\varphi)^2\right], \quad (11.2)$$

with $A_\mu = A_\mu^a\tau^a/2$, $D_\mu = \partial_\mu - igA_\mu$, $G_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu - ig[A_\mu, A_\nu]$, so that $-\frac{1}{2}\text{Tr}(G_{\mu\nu}G^{\mu\nu}) = -\frac{1}{4}G_{\mu\nu}^a G^{a\mu\nu}$. The energy is

$$E(A_\mu^a, \varphi) = \int d^3x \left[\frac{1}{2}G_{0i}^a G_{0i}^a + \frac{1}{4}G_{ij}^a G_{ij}^a + (D_0\varphi)^\dagger(D_0\varphi) + (D_i\varphi)^\dagger(D_i\varphi) + V(\varphi^\dagger\varphi) \right]. \quad (11.3)$$

In the ground state $G_{\mu\nu}^a = 0$, so A_μ is a pure gauge,

$$A_\mu(x) = \frac{i}{g}\omega(x)\partial_\mu\omega^{-1}(x), \quad \omega(x) = e^{i\tau^a u^a(x)/2} \in SU(2), \quad (11.4)$$

and the scalar field is covariantly constant, $D_\mu\varphi = 0 \Rightarrow \varphi(x) = \omega(x)\varphi^{(v)}$. The gauge equivalent vacua are therefore represented by

$$A_\mu^a = 0, \quad \varphi^{(v)} = \begin{pmatrix} 0 \\ \varphi_0/\sqrt{2} \end{pmatrix}, \quad \varphi_0 = \frac{\mu}{\sqrt{\lambda}}. \quad (11.5)$$

Fluctuations and unitary gauge. Including fluctuations,

$$\varphi(x) = \begin{pmatrix} \xi_1(x) + i\xi_2(x) \\ \frac{\varphi_0}{\sqrt{2}} + \chi(x) + i\xi_3(x) \end{pmatrix} \simeq \exp\left(\frac{i\tau^a \xi^a(x)}{\varphi_0/\sqrt{2}}\right) \begin{pmatrix} 0 \\ \frac{\varphi_0}{\sqrt{2}} + \chi(x) \end{pmatrix} \quad (11.6)$$

to linear order (with a relabelling of the ξ^a). The three ξ^a have the form of a gauge transformation, and can be removed by an $SU(2)$ gauge transformation; this choice is the *unitary gauge*. In it, the Goldstone modes ξ^a are gauged away, and through the Higgs mechanism they are absorbed by the gauge bosons, which become massive.

$SU(2)$ has three generators, i.e. three gauge bosons, initially massless. In the vacuum $\varphi^{(v)}$ all three generators are broken (no gauge symmetry remains), so by the Goldstone counting there would be three Goldstone modes (ξ_1, ξ_2, ξ_3). All three gauge bosons ($A_\mu^1, A_\mu^2, A_\mu^3$) become massive by eating them, and the ξ^a vanish from the spectrum. Using $\varphi^{(v)\dagger} \tau^a \tau^b \varphi^{(v)} + (a \leftrightarrow b) = \delta^{ab} \varphi_0^2$, after SSB in the unitary gauge,

$$\mathcal{L} = -\frac{1}{4} G_{\mu\nu}^a G^{a\mu\nu} + \frac{g^2 \varphi_0^2}{8} A_\mu^a A^{a\mu} + \frac{1}{2} (\partial_\mu \chi)^2 - \mu^2 \chi^2 + \dots, \quad (11.7)$$

$$\boxed{m_A = \frac{g\varphi_0}{2}, \quad m_\chi = \sqrt{2}\mu.} \quad (11.8)$$

Counting: $3 \times 2 + 4 = 10$ degrees of freedom before, $3 \times 3 + 1 = 10$ after.

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